

NTRK3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK3 gene fusions from the msk_impact_50k_2026 study, covering 58 fusion events. 55/57 fusions (96.5%) were in-frame. 55/57 fusions (96.5%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0695489$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 58 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was ETV6, observed in 55 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 54/55 (98.2%) of in-frame fusions retained the domain, $p=0.0695489$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Domain-disruption analysis was skipped: NTRK3 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 57 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (exon 5/6 boundary; permutation-corrected $p=0.036963$, statistically significant at $\alpha=0.05$). This is -5 aa from the nearest configured domain boundary at 160 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=55$) and "not_retained" ($n=2$). For Frame_status, retained: 54/54 (100.0%, 95% CI 93.4%-100.0%); not_retained: 1/2 (50.0%, 95% CI 9.5%-90.5%). Welch's t-test comparing Tumor_variant_count between groups gave means of 74.1455 vs 7, $p=5.97753e-12$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for NTRK3 in this cohort; expression-association analysis was skipped.

Joint partner

Joint partner reported: NTRK3 has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: NTRK3 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (18 row(s) in window_scan, 4 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ETV6	55	0.9482758620689655	0.30004340774793187	None	0.0	0.5613244116887826	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.5195325646581886	1
DOT1L	1	0.017241379310344827	0.30004340774793187	None	1.0	0.5613244116887826	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3578946336237058	2
VPS39	1	0.017241379310344827	0.30004340774793187	None	None	0.5613244116887826	confidence_certainty, domain_retention, recurrence	0.2097164721522533	3
ZNF710	1	0.017241379310344827	0.30004340774793187	None	0.11159235668789813	0.5613244116887826	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.19131820050268672	4

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
155	0	55	1	1	0.0	0.03508771929824562	1.4548448600085102
466	14	41	2	0	0.0	0.07518796992481205	1.1238516409670856
472	15	40	2	0	0.0	0.08521303258145364	1.069493978644493

domain_retention tables

frame_domain_contingency_table

54	1
1	1

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	57	55	0	2	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
14	55	58	0.9482758620689655

frequency tables

Partner_gene_counts

Partner_gene	Event_count
DOT1L	1
ETV6	55
VPS39	1
ZNF710	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
266	466	200	15	42	14	41	0.34146341463414637	0.46052631578947384	0.3367455479305156	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (15 total)
272	472	200	16	41	15	40	0.375	0.486215538847118	0.313171165756522	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total)
366	466	100	15	42	14	41	0.34146341463414637	0.46052631578947384	0.3367455479305156	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (15 total)
372	472	100	16	41	15	40	0.375	0.486215538847118	0.313171165756522	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
416	466	50	15	42	14	41	0.34146341463414637	0.46052631578947384	0.3367455479305156	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (15 total)
422	472	50	16	41	15	40	0.375	0.486215538847118	0.313171165756522	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total)
441	466	25	15	42	14	41	0.34146341463414637	0.46052631578947384	0.3367455479305156	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (15 total)
447	472	25	16	41	15	40	0.375	0.486215538847118	0.313171165756522	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total)
466	491	25	16	41	15	40	0.375	0.486215538847118	0.313171165756522	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total)
466	516	50	16	41	15	40	0.375	0.486215538847118	0.313171165756522	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total)
472	572	100	41	16	41	14	None	0.07518796992481204	1.1238516409670858	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (41 total)
472	672	200	41	16	41	14	None	0.07518796992481204	1.1238516409670858	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (41 total)
479	529	50	40	17	40	15	None	0.08521303258145364	1.069493978644493	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (40 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
504	529	25	40	17	40	15	None	0.08521303258145364	1.069493978644493	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (40 total)
529	554	25	40	17	40	15	None	0.08521303258145364	1.069493978644493	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (40 total)
529	579	50	40	17	40	15	None	0.08521303258145364	1.069493978644493	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (40 total)
529	629	100	40	17	40	15	None	0.08521303258145364	1.069493978644493	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (40 total)
529	729	200	40	17	40	15	None	0.08521303258145364	1.069493978644493	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (40 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
472	572	100	41	16	41	14	None	0.07518796992481204	1.1238516409670858	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (41 total)
504	529	25	40	17	40	15	None	0.08521303258145364	1.069493978644493	EVT-P-0002671-T01-IM3-30, EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45, EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56, ... (40 total)
441	466	25	15	42	14	41	0.34146341463414637	0.46052631578947384	0.3367455479305156	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (15 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
447	472	25	16	41	15	40	0.375	0.486215538847118	0.313171165756522	EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0013664-T02-IM5-1	P-0013664-T02-IM5		Soft Tissue Sarcoma	FIBS	ETV6-NTRK3	ETV6	in-frame	three_prime	88558191	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q26.1)(chr12:g.12032077::chr15:g.88558191) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0061265-T01-IM7-2	P-0061265-T01-IM7		Salivary Gland Cancer	HNMC	ETV6-NTRK3	ETV6	in-frame	three_prime	88554122	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12033175::chr15:g.88554122) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058408-T01-IM6-3	P-0058408-T01-IM6		Colorectal Cancer	COAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88492156	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12035760::chr15:g.88492156) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0056156-T02-IM7-4	P-0056156-T02-IM7		Thyroid Cancer	THPA	ETV6-NTRK3	ETV6	in-frame	three_prime	88624274	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12011957::chr15:g.88624274) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055703-T01-IM6-5	P-0055703-T01-IM6		Cancer of Unknown Primary	CUP	ETV6-NTRK3	ETV6	in-frame	three_prime	88505980	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12024785::chr15:g.88505980) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0042503-T01-IM6-6	P-0042503-T01-IM6		Colorectal Cancer	COAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88556067	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032802::chr15:g.88556067) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040355-T01-IM6-7	P-0040355-T01-IM6		Colorectal Cancer	COAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88497191	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12036111::chr15:g.88497191) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0037610-T01-IM6-8	P-0037610-T01-IM6		Thyroid Cancer	THPA	ETV6-NTRK3	ETV6	in-frame	three_prime	88594532	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) rearrangement: t(12;15)(p13.2;q25.3)(chr12:g.12014663::chr15:g.88594532) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035767-T06-IM7-9	P-0035767-T06-IM7		Pancreatic Cancer	PAAC	ETV6-NTRK3	ETV6	in-frame	three_prime	88505787	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0035767-T04-IM6-10	P-0035767-T04-IM6		Pancreatic Cancer	PAAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88505787	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0033921-T05-IM7-11	P-0033921-T05-IM7		Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0033921-T04-IM6-12	P-0033921-T04-IM6		Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0033921-T02-IM6-13	P-0033921-T02-IM6		Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0032148-T04-IM6-14	P-0032148-T04-IM6		Salivary Gland Cancer	HNMC	ETV6-NTRK3	ETV6	in-frame	three_prime	88522437	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12037217::chr15:g.88522437) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064964-T01-IM7-15	P-0064964-T01-IM7		Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88564682	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032310::chr15:g.88564682) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0013664-T04-IM6-16	P-0013664-T04-IM6		Soft Tissue Sarcoma	FIBS	ETV6-NTRK3	ETV6	in-frame	three_prime	88558191	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032077::chr15:g.88558191) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058230-T02-IM7-19	P-0058230-T02-IM7		Pancreatic Cancer	PAAD	NTRK3-DOT1L	DOT1L	out-of-frame	five_prime	88683244	5	155	True	154	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		NTRK3 (NM_001012338) - DOT1L (NM_032482) rearrangement: t(15;19)(q25.3;p13.3)(chr15:g.88683244::chr19:g.2227604) Protein Fusion: out of frame {NTRK3:DOT1L}	NA
EVT-P-0013196-T03-IM6-20	P-0013196-T03-IM6		Breast Cancer	BNNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T06-IM6-21	P-0002978-T06-IM6		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0010668-T01-IM5-22	P-0010668-T01-IM5		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88513282	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029109::chr15:g.88513282) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014872-T02-IM6-23	P-0014872-T02-IM6		Soft Tissue Sarcoma	IMT	NTRK3-ETV6	ETV6	in-frame	three_prime	88548029	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0014872-T03-IM6-24	P-0014872-T03-IM6		Soft Tissue Sarcoma	IMT	NTRK3-ETV6	ETV6	in-frame	three_prime	88548029	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013196-T04-IM6-25	P-0013196-T04-IM6		Breast Cancer	BRCNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0016507-T01-IM6-26	P-0016507-T01-IM6		Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016507-T02-IM6-27	P-0016507-T02-IM6		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-14 fused with NTRK3 exons 6-20 including the kinase domain): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-016507-T03-IM6-28	P-0016507-T03-IM6		Salivary Gland Cancer	HNMC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20: t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-016507-T04-IM6-29	P-0016507-T04-IM6		Salivary Gland Cancer	HNMC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002671-T01-IM3-30	P-0002671-T01-IM3		Colorectal Cancer	COAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88505784	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0018060-T01-IM6-31	P-0018060-T01-IM6		Salivary Gland Cancer	HNMSC	NTRK3-ETV6	ETV6	in-frame	three_prime	88514531	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-021974-T01-IM6-32	P-0021974-T01-IM6		Salivary Gland Cancer	SAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88550500	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028417::chr15:g.88550500) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-013196-T02-IM6-33	P-0013196-T02-IM6		Breast Cancer	BRCNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002507-T01-IM6-34	P-0025507-T01-IM6		Breast Cancer	IDC	NTRK3-ETV6	ETV6	in-frame	three_prime	88544573	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12024078::chr15:g.88544573) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0025987-T01-IM6-35	P-0025987-T01-IM6		Miscellaneous Brain Tumor	PRNET	NTRK3-ETV6	ETV6	in-frame	three_prime	88613114	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12028942::chr15:g.88613114) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010562-T01-IM5-36	P-0010562-T01-IM5		Salivary Gland Cancer	SDCA	NTRK3-ETV6	ETV6	in-frame	three_prime	88581658	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 14-20): t(12;15)(p13.2;q25.1)(chr12:g.12028903::chr15:g.88581658) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0026390-T01-IM6-37	P-0026390-T01-IM6		Thyroid Cancer	THPA	NTRK3-ETV6	ETV6	in-frame	three_prime	88651131	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863::chr15:g.88651131) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026696-T01-IM6-38	P-0026696-T01-IM6		Thyroid Cancer	THPA	NTRK3-ETV6	ETV6	in-frame	three_prime	88657925	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12007967::chr15:g.88657925) Protein Fusion: in frame {ETV6:N TRK3}	NA
EVT-P-0027006-T01-IM6-39	P-0027006-T01-IM6		Soft Tissue Sarcoma	SARCN OS	NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:N TRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-027006-T03-IM6-40	P-0027006-T03-I M6		Soft Tissue Sarcoma	FIBS	NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-027006-T04-IM6-41	P-0027006-T04-I M6		Soft Tissue Sarcoma	FIBS	NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T07-IM6-42	P-0002978-T07-IM6		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0002978-T02-IM5-43	P-0002978-T02-IM5		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026286-T01-IM6-44	P-0026286-T01-IM6		Salivary Gland Cancer	SACA	NTRK3-ETV6	ETV6	in-frame	three_prime	88611476	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12030395::chr15:g.88611476) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0002978-T03-IM5-45	P-0002978-T03-IM5		Salivary Gland Cancer	HNMA5C	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 5 fused to NTRK3 exons 15 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010264-T01-IM5-46	P-0010264-T01-IM5		Pancreatic Cancer	PAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88588255	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused in frame with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12014962::chr15:g.88588255) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007911-T02-IM5-47	P-0007911-T02-IM5		Thyroid Cancer	THPD	NTRK3-ETV6	ETV6	in-frame	three_prime	88592430	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12007142::chr15:g.88592430) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0033921-T03-IM6-48	P-0033921-T03-IM6		Breast Cancer	BRCNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-007772-T03-IM5-49	P-007772-T03-IM5		Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88600844	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0007772-T02-IM5-50	P-0007772-T02-IM5		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	five_prime	88600844	12	466	True	465	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (NTRK3-ETV6)	NA
EVT-P-0035552-T01-IM6-51	P-0035552-T01-IM6		Pancreatic Cancer	PAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88651251	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12020963::chr15:g.88651251) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035767-T01-IM6-52	P-0035767-T01-IM6		Pancreatic Cancer	PANET	NTRK3-ETV6	ETV6	in-frame	three_prime	88505787	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-007772-T01-IM5-53	P-007772-T01-IM5		Salivary Gland Cancer	HNMC	NTRK3-ETV6	ETV6	in-frame	three_prime	88600844	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA

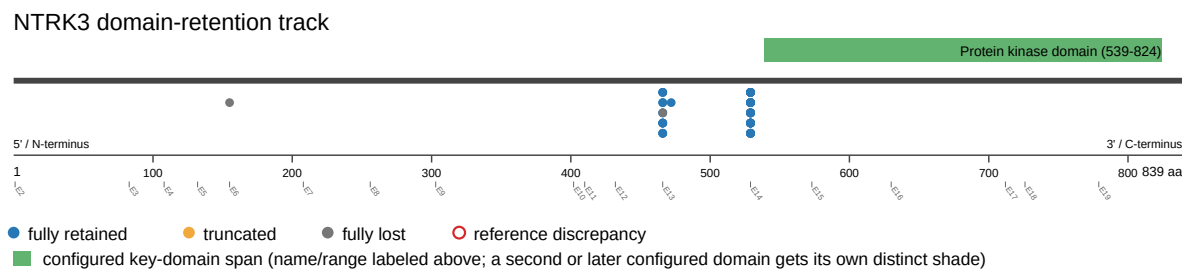
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006480-T01-IM5-54	P-0006480-T01-IM5		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88498145	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548::chr15:g.88498145) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0002978-T04-IM5-55	P-0002978-T04-IM5		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T05-IM6-56	P-0002978-T05-IM6		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0004737-T01-IM5-57	P-0004737-T01-IM5		Pancreatic Cancer	PAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88632786	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 13 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12006809::chr15:g.88632786) Protein fusion: in frame (ETV6-NTRK3)	NA

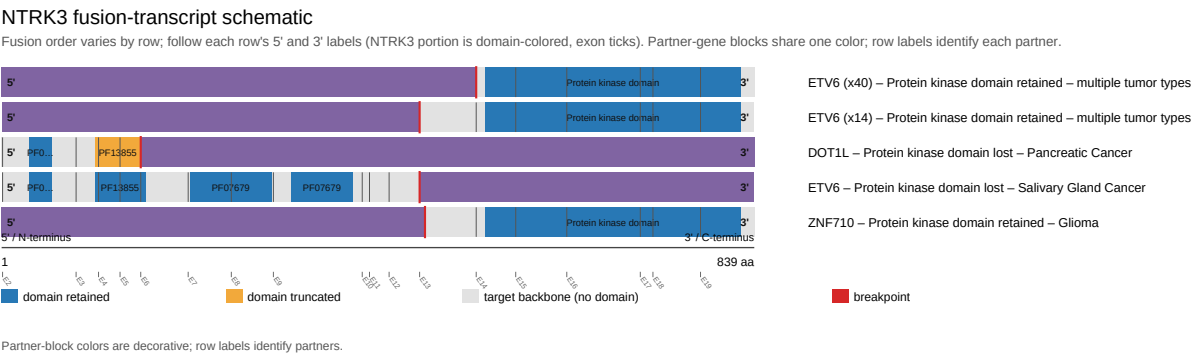
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-023257-T01-IM6-58	P-0023257-T01-IM6		Salivary Gland Cancer	HNMAC	NTRK3-ETV6	ETV6	in-frame	three_prime	88662190	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998::chr15:g.88662190) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-010556-T01-IM5-70	P-0010556-T01-IM5		Glioma	GB	NTRK3-ZNF710	ZNF710	unknown	three_prime	88576258	13	472	False	472	True	exact match	retained	1.0	False	Protein kinase domain			NTRK3 (NM_001012338) - ZNF710 (NM_198526) rearrangement: c.-28-10227:ZNF710_c.1415:NTRK3inv Protein fusion: mid-exon (ZNF710-NTRK3)	NA

Figures

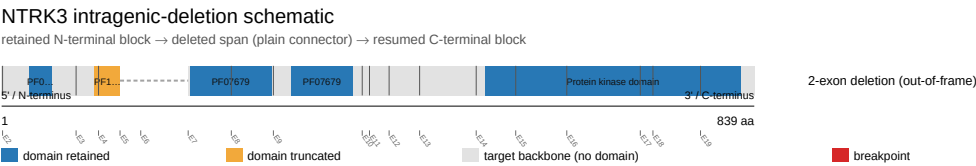
domain retention outliers



fusion schematic



intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

